

ETABS input guide - Model A (as-built flat slab) and Model B (proposed beam-slab)

State Audit Office building, Phase 2 comparison with the OpenSees models (version 2.0). Follow this guide to build the two ETABS models so that they describe exactly the same structure as the OpenSees models, run the same analyses, and paste the results into the comparison templates.

Workflow

Step	Where	What
1	ETABS	Build Model A and Model B with the inputs in Sections 1-8 (same grid, columns, core; only the floor framing differs).
2	ETABS	Run the gravity, modal and response-spectrum cases (Section 9). Optional: the pushover (Section 11).
3	ETABS -> Excel	Export the tables listed in Section 10 and copy the values into the CSV templates in phase2_opensees_2.0/etabs_comparison/templates/; save them in etabs_comparison/filled/ with the same file names.
4	OpenSees folder	Run python3 etabs_comparison/compare_with_etabs.py. It writes a table of ETABS vs OpenSees with the difference in % and PASS / CHECK flags, plus overlay figures, into etabs_comparison/output/.

What the comparison checks. It is an **elastic** comparison (the standard ETABS Phase 1 analyses): seismic weight, periods and mass participation, response-spectrum base shear and its split between core and columns, storey shears, storey drifts, roof displacement and gravity axial forces. A pushover comparison is optional (Section 11) because ETABS hinges cannot reproduce punching or the wall failure rules used in OpenSees.

Two conditions are compared: **REF** = the original design (no deficiencies) and **D1D2D3D4** = the documented as-built condition. Start with REF.

1. Units, code and general settings

Item	Setting
Units	kN, m, C (consistent: stresses in kN/m ² ; enter MPa x 1000)
Concrete code	ACI 318-19 (for section properties only; no design check needed)
P-Delta	Off for the elastic comparison (the OpenSees modal / RSA model has no initial P-Delta). On (non-iterative, based on gravity) only for the optional pushover.
Diaphragms	One rigid diaphragm per floor (D1 ... D33) assigned to all slab/joint objects of the floor
Base	Fixed supports (all 6 DOF) at the base of every column and wall at level 0 (no soil springs)
Auto mesh	Walls: 1.0 m max; slabs: 1.0 m max (rectangular mesh); add walls/columns as mesh points

2. Grid and storeys (both models)

Grid	Lines (m)
X lines A-F	0, 5.5, 14.75, 24.25, 33.5, 39
Y lines 1-6	0, 5.5, 14.75, 24.25, 33.5, 39
Plan	39 m x 39 m; the core is at the rear (y = 39 m) edge, centred in x

Storey	Height (m)	Elevation of floor (m)
Base	-	0.0
Story 1 (ground)	6.000	6.000
Story 2 ... Story 33 (32 typical)	4.09375	10.094 ... 137.000

Total height 137.0 m. Use 'Similar to' so all 33 floors share one plan.

3. Materials

Isotropic concrete, Poisson 0.2, unit weight 24 kN/m³, E = 4700 sqrt(f_c) MPa. Rebar SD50 f_y = 490 MPa, SD40 f_y = 390 MPa, E_s = 200,000 MPa (rebar only matters for the pushover).

Material name	f'c (MPa)	E (MPa)	Used for
C49-WALL	49.0	32,900	Core walls + link beams, storeys 1-16 (REF)
C40-WALL	40.0	29,725	Core walls + link beams, storeys 17-33 (REF)
C34-WALL-D1	34.3	27,526	Walls storeys 1-16, as-built condition (D1 = 0.70 f'c)
C28-WALL-D1	28.0	24,870	Walls storeys 17-33, as-built condition (D1)
C55-COL	55.0	34,856	Columns storeys 1-11
C45-COL	45.0	31,529	Columns storeys 12-22
C35	35.0	27,806	Columns storeys 23-33, all slabs, beams (Model B)

4. Columns (both models)

26 columns per floor - **no columns at the four plan corners** and none inside the core.

#	x (m)	y (m)	#	x (m)	y (m)	#	x (m)	y (m)
1	0	5.5	2	0	14.75	3	0	24.25
4	0	33.5	5	5.5	0	6	5.5	5.5
7	5.5	14.75	8	5.5	24.25	9	5.5	33.5
10	5.5	39	11	14.75	0	12	14.75	5.5
13	14.75	14.75	14	24.25	0	15	24.25	5.5
16	24.25	14.75	17	33.5	0	18	33.5	5.5
19	33.5	14.75	20	33.5	24.25	21	33.5	33.5
22	33.5	39	23	39	5.5	24	39	14.75
25	39	24.25	26	39	33.5			

Storeys	Section	Concrete	Long. steel ratio (pushover only)	Stiffness modifiers
1-3	1400 x 1400 rectangular	C55-COL	2.0 %	I22 = I33 = 0.70; J = 0.20
4-12	1200 x 1200 rectangular	C55-COL (1-11), C45-COL (12)	2.0 % (<=6), 1.5 %	same
13-20	1000 x 1000 rectangular	C45-COL	1.5 % (<=15), 1.0 %	same
21-28	900 x 900 rectangular	C45-COL (<=22), C35	1.0 %	same
29-33	800 diameter circular	C35	1.0 %	same

5. Core walls and link beams (both models)

The core is a 17 x 17 m box (wall centrelines x = 11 to 28 m, y = 22 to 39 m) with a cross wall at y = 30.5 m. Each side wall (x = 11 and x = 28) has **two 1.5 m openings**, at y = 25.5-27.0 and y = 34.0-35.5 m, full storey height minus the link beam. The openings split the core into three **piers**, which must be given ETABS **pier labels R, M, F** (the OpenSees results are reported per pier):

Pier label	Wall pieces (centreline coordinates)	Shape
F (front)	Front wall y = 22.0, x 11->28; side walls x = 11 and x = 28 from y = 22.0 to 25.5	C
M (middle)	Cross wall y = 30.5, x 11->28; side walls x = 11 and x = 28 from y = 27.0 to 34.0	I
R (rear)	Rear wall y = 39.0, x 11->28 (building rear face); side walls x = 11 and 28 from y = 35.5 to 39.0	C

Item	Model input
Wall element	Shell-thin wall, thickness 300 mm (REF); 250 mm for the as-built condition (D3)
Wall concrete	C49-WALL storeys 1-16, C40-WALL storeys 17-33 (REF); C34 / C28 for D1
Wall stiffness modifiers	f11 = f22 = f12 = 0.50 (0.5 EI and 0.5 GA in OpenSees); m11 = m22 = m12 = 0.25; weight 1.0
Link beams (4 per floor)	Across each 1.5 m opening at every floor level: 300 wide (= wall t; 250 for D3) x 500 deep . Model as frame elements (or spandrels) joined to the wall edges; label them as spandrels S1-S4.
Link beam modifiers	I33 = 0.20 , I22 = 0.20, J = 0.20 (ACI/TBI effective stiffness); shear area 1.0
Link beam reinforcement (pushover)	3300 mm ² top and bottom, SD50; D2: embedment = 0.5 of the development length

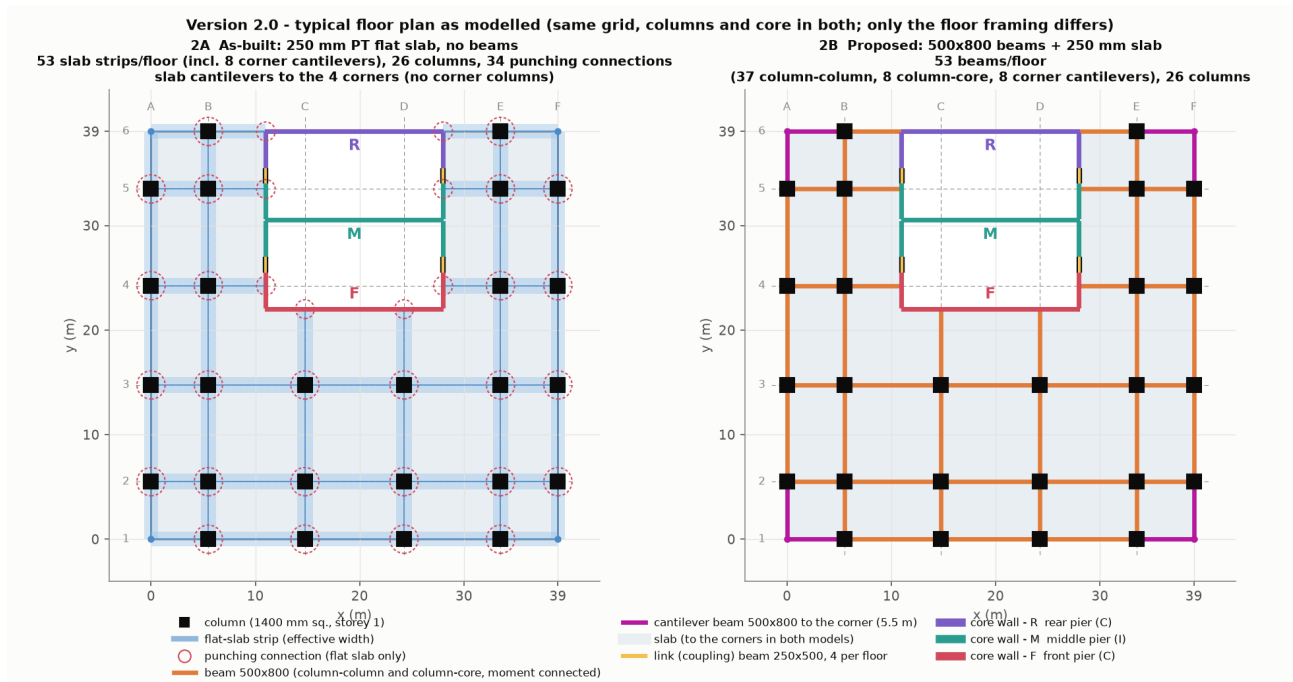


Figure 1. Typical floor plans exactly as modelled in OpenSees v2.0 (Model A left, Model B right). Columns black, core piers R / M / F coloured, link beams yellow, corner cantilevers magenta.

6. Floor framing - the only difference between the models

6A. Model A - as-built post-tensioned flat slab

Item	Model input
Slab	250 mm shell-thin slab, C35, covering the whole 39 x 39 m floor to the corners (the corners cantilever 5.5 m from the edge columns). No beams, no drop panels.
Inside the core	No structural slab (shafts). Draw a 'None' (null) area over the core interior to carry the landing load (Section 7).
Slab stiffness modifiers	Bending $m_{11} = m_{22} = m_{12} = \mathbf{0.25}$ (cracked PT flat plate); membrane 1.0 (rigid diaphragm anyway).
Post-tensioning	Do not model tendons for the elastic comparison (they do not change elastic stiffness). OpenSees uses 1.5 MPa average precompression only in the punching strength.
Alternative (exact match)	OpenSees represents the slab as effective-width strips along the column lines: width $2c_c + l_1/3$ (interior) or $c_1 + l_1/6$ (edge), 250 mm deep, 0.5 EI. Shell slabs with $m = 0.25$ are the normal ETABS practice and should give periods within about 10 %.

6B. Model B - proposed beam-slab

Item	Model input
Slab	250 mm shell-thin slab, C35, whole floor to the corners (except inside the core); bending $m = 0.25$
Beams	500 x 800 rectangular frame sections, C35, on every column line between supports (column-column and column-core wall) - 45 per floor - plus 8 corner cantilevers (5.5 m from the nearest edge column to each corner, along both edges): 53 beams per floor
Beam insertion point	Top centre (8) so the slab sits on top of the beams
Beam stiffness modifiers	$I_{33} = \mathbf{0.74}$ if the beam is a plain 500 x 800 rectangle ($= 0.35 \times I_{gg}$ of the T-section with 6t flanges / I_{gg} of the rectangle; OpenSees uses $0.35 I_{g,T}$). Use 0.35 if you define a T-section. $I_{22} = 0.35$, $J = 0.20$.
Beam end connections	No releases - all beams, including the 8 framing into the core walls, are fully moment connected. At wall ends add a short rigid link or mesh the wall so the beam frames into wall nodes.
Beam reinforcement (pushover)	Top / bottom A_s (mm^2), perimeter interior beams: Z1: 4682/4297 5233/3275; Z2: 5023/4562 6108/3414; Z3: 4970/3741 6449/3224; Z4: 4214/2206 6435/3217 (zones Z1 storeys 1-8, Z2 9-16, Z3 17-24, Z4 25-33)

7. Load patterns and mass source

Load pattern	Type	Design values (elastic comparison)	Event values (pushover gravity)
DEAD	Dead	Self-weight multiplier 1.0 (slab $0.25 \times 24 = 6.0$ kPa; beams, walls, columns automatic)	same
SDL	Super dead	1.50 kPa on all slab areas	0.75 kPa
LIVE	Live	2.50 kPa on all slab areas	0.50 kPa
FACADE	Super dead	Line load on the full perimeter (156 m) = 1.50 kPa x storey height: 9.00 kN/m storey 1, 6.14 kN/m typical (half each side)	0.75 kPa x storey height
CORE (null area)	Super dead	$0.30 \times (6.0 + 1.5 + 2.5) = 3.00$ kPa over the core interior (landings, stairs)	$0.30 \times (6.0 + 0.75 + 0.5) = 2.17$ kPa

Mass source: 'From loads' - DEAD 1.0 + SDL 1.0 + FACADE 1.0 + CORE 1.0 + **LIVE 1.0** (OpenSees takes the full gravity load as mass; if you prefer $0.25 \times \text{LIVE}$ note that the weight will be about 6-7 % lower and adjust). Lumped at stories, include lateral mass only, no vertical mass. Target seismic weight (REF, design loads): Model A 632 MN, Model B 715 MN.

8. Response spectrum function (DPT 1301/1302-61, design level)

T (s)	0	0.2	0.5	0.8	1	1.5	2
Sa (g)	0.0800	0.1333	0.1800	0.2000	0.2000	0.1933	0.1733
T (s)	2.5	3	4	5	6	8	10
Sa (g)	0.1467	0.1200	0.0867	0.0667	0.0567	0.0400	0.0300

Define as a 'From file / user' function with damping 5 %. These are the design ($2/3 \times \text{MCE}$) spectral accelerations in g, before dividing by R / I .

9. Load cases to run

Case	Type	Settings
GRAV	Linear static	DEAD 1.0 + SDL 1.0 + LIVE 1.0 + FACADE 1.0 + CORE 1.0 (design values) - gives the pier and column axial forces
MODAL	Eigen	12 modes, no P-Delta, mass from the mass source
RSX	Response spectrum	U1, function DPT, scale factor = $9.81 \times I / R = 9.81 \times 1.25 / 5 = 2.4525$; modal combination CQC , damping 5 %; directional combination: single direction; no accidental eccentricity ; do not scale to ELF
RSY	Response spectrum	Same in U2

Do **not** scale the response-spectrum results up to 85 % of ELF: the comparison uses the unscaled values (OpenSees RSA / ELF = 0.34-0.44). The ELF base shear ($C_s = 0.0335$, T capped at 1.4 Ta = 2.74 s) is reported separately.

10. What to export from ETABS and where it goes

Templates (one set per model and condition): ETABS_summary_<A|B>_<REF|D1D2D3D4>.csv and ETABS_storeys_<A|B>_<REF|D1D2D3D4>.csv. Each row already shows the OpenSees value; type the ETABS value in the empty column. Leave a cell empty if you do not have the value.

Template key(s)	ETABS table (Display > Show Tables)	What to copy
W	Mass Summary by Story (or Base Reactions of GRAV)	Total mass x 9.81, or total vertical reaction of GRAV (kN)
T1 ... T6	Modal Periods and Frequencies	Period of modes 1-6 (s)
UX1 ... RZ6, SumUX/UY/RZ	Modal Participating Mass Ratios	UX, UY, RZ of modes 1-6 and the cumulative sums at mode 12, in % (ETABS gives fractions: multiply by 100)
V_RSA_X, V_RSA_Y	Base Reactions	FX of RSX and FY of RSY (kN)
Vcore_RSA_X / _Y	Pier Forces, Story1, Bottom	Sum of V2 (in-plane) of piers R, M, F for RSX / RSY - or the X / Y components if you report in global axes
roof_RSA_X / _Y	Joint Displacements, a joint at the plan centre (19.5, 19.5) of the roof	UX for RSX, UY for RSY, in mm
P1_R, P1_M, P1_F, P1_cols	Pier Forces, Story1, Bottom, case GRAV; Column Forces Story1	Axial P of each pier; sum of P of the 26 columns (kN, compression positive)
Storeys sheet: shear X / Y	Story Forces, Location = Bottom, RSX / RSY	VX for RSX and VY for RSY (kN), storey 1-33
Storeys sheet: drift centre	Joint Drifts of the plan-centre joint (19.5, 19.5) of each floor	Drift X for RSX, drift Y for RSY, in % (= ratio x 100)
Storeys sheet: drift max	Story Drifts (max over the floor)	Drift X for RSX, drift Y for RSY, in %

Then save the filled templates into etabs_comparison/filled/ and run python3 etabs_comparison/compare_with_etabs.py from the phase2_opensees_2.0 folder. Acceptance used: weight +/-5 %, periods and mass ratios +/-10 %, base shears +/-15 %, roof displacement +/-20 %, gravity forces +/-10 %. Typical reasons for a CHECK: a different slab stiffness modifier, a missed link beam or opening, beams released at the core, LIVE taken as 0.25 in the mass source, or P-Delta switched on.

11. Optional - pushover in ETABS

Item	Setting (to match the OpenSees pushover)
Gravity first	Nonlinear static case PGRAV: DEAD 1.0 + SDL 1.0 + LIVE 1.0 + FACADE 1.0 + CORE 1.0 with the event values of Section 7; P-Delta on
Lateral load	User load pattern: at the plan centre of each floor a force F_i proportional to $W_i z_i^2$ (W_i floor weight, z_i floor elevation), in X (PUSHX) or Y (PUSHY)
Case	Nonlinear static, start from PGRAV, displacement control at the roof plan-centre joint, up to 3 % of 137 m = 4.11 m, save multiple states
Hinges - columns	Auto P-M2-M3, ASCE 41-13 Table 10-8, at both ends (relative distance 0.05 / 0.95)
Hinges - walls	Fibre P-M3 hinges at every storey of piers R, M, F (or at least storeys 1-4), confined ends only for REF (D4: unconfined)
Hinges - link beams	Auto V2 / M3, ASCE 41-13 Table 10-19, conventional (non-diagonal) reinforcement; for D2 cap the moment at 0.5 x the full value

Item	Setting (to match the OpenSees pushover)
Hinges - beams (B)	Auto M3, ASCE 41-13 Table 10-7, at both ends including at the core walls
Not reproducible in ETABS	Flat-slab punching (ends the OpenSees Model A curve at about 1 % roof drift in X) and the wall axial-failure removal; expect ETABS curves to continue further
Export	Pushover Curve (base shear vs monitored displacement) -> templates ETABS_pushover_<A B>_<X Y>.csv (roof_displacement_m, base_shear_kN)

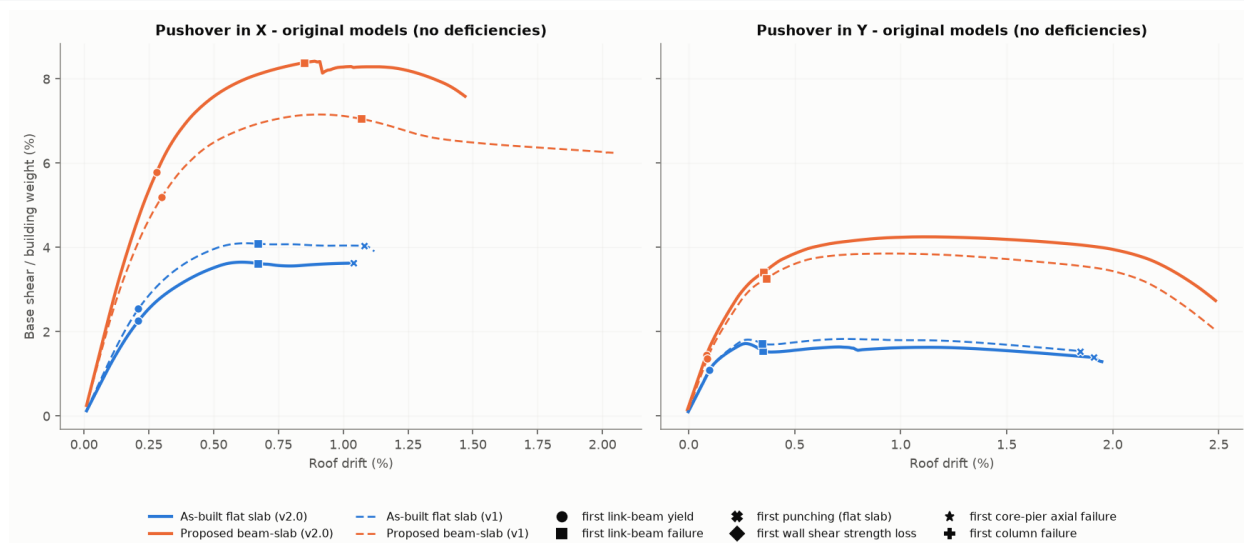


Figure 2. OpenSees pushover curves of the original models (base shear / weight vs roof drift); the comparison script plots your ETABS curves on top (in MN vs m).

12. As-built condition (D1-D4) - what to change in ETABS

Deficiency	Change in ETABS	Effect on the elastic comparison
D1 low wall concrete (finding 2)	Walls and link beams use C34-WALL-D1 / C28-WALL-D1 (0.70 f _c)	Lower wall stiffness: longer periods
D2 short link-beam embedment (finding 4)	No change for elastic analysis; pushover link-beam hinge capped at 0.5 x moment	None (elastic)
D3 walls 300 -> 250 mm (noted reduction)	Wall and link-beam thickness 250 mm	Lower stiffness and weight
D4 non-compliant design (finding 3)	No change for elastic analysis (reinforcement only); pushover: unconfined wall hinges, wall horizontal steel R 0.33 %, M 0.39 %, F 0.35 % at storeys 1-8	None (elastic)

Compare the ETABS as-built condition against the OpenSees file E0_opensees_elastic_D1D2D3D4.json (templates *_D1D2D3D4.csv).

13. OpenSees target values (REF, design loads) - quick check while building

Quantity	Model A	Model B
Seismic weight	632 MN	715 MN
T1 / T2 / T3	7.65 / 5.94 / 3.82 s	4.96 / 4.80 / 2.72 s
Mode 1 UX / RZ	30 / 40 %	33 / 41 %
Mode 2 UY	72 %	74 %
RSA base shear X / Y	7.2 / 7.8 MN	10.7 / 10.5 MN
Roof displacement X / Y	123 / 180 mm	84 / 141 mm
Storey-1 pier axial R / M / F	41.7 / 63.6 / 62.8 MN	47.4 / 72.5 / 73.7 MN

If ETABS mode 1 is a pure X translation instead of coupled X + torsion (about 30 % UX + 40 % RZ), the core position or the diaphragm assignment differs between the models - check those first.